

# AANCHAL RATHI

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## Education

- **VIT Bhopal University, Bhopal** Sep 2023 – Present  
Bachelor of Technology in Computer Science (AI & ML Specialization)  
CGPA: 8.95 / 10
- **DAV Public School** 2019 – 2023  
12th Grade: 83.8% (2023)  
10th Grade: 94% (2021)

## Technical Skills

- **Languages:** C++, Java, Python, JavaScript
- **Cloud Platforms:** Microsoft Azure (Certified), AWS
- **Machine Learning:** Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, GNN, OpenCV
- **Web and Backend:** RESTful APIs, JavaScript-based integration of ML models
- **Databases:** MySQL, MongoDB (basic)
- **Other Skills:** Problem Solving, Team Collaboration, Agile Basics

## Projects

### Power Consumption Analysis using Machine Learning May 2025 – Jul 2025

*Tech Stack: Python, Pandas, Scikit-learn, Matplotlib, Flask, HTML, CSS*

- Built an analytics platform using smart meter and weather data to study household energy usage.
- Cleaned and prepared UCI dataset; trained models to detect inefficiencies like standby drains and HVAC overuse.
- Tuned hyperparameters, evaluated using accuracy and F1-score, and deployed via Flask dashboard.

### Crop Recommendation using Machine Learning Feb 2025 – Apr 2025

*Tech Stack: Python, Pandas, NumPy, Scikit-learn*

- Developed a model using soil nutrients, temperature, humidity, pH, and rainfall data.
- Performed EDA, feature selection, and trained Decision Tree, KNN, Naive Bayes, and SVM models.
- Achieved 95% accuracy with Decision Tree; supports data-driven crop recommendations.

### Drug Discovery using Graph Neural Networks Aug 2024 – Oct 2024

*Tech Stack: Python, PyTorch Geometric, Scikit-learn, SMILES*

- Built ML pipeline by converting SMILES into molecular graphs with engineered atom and bond features.
- Compared Random Classifier with GNN; improved accuracy from 75% to 87.9%.
- Applied SHAP and graph activations for interpretability.

## Achievements & Profiles

- Solved 500+ DSA problems across Codeforces, LeetCode and GeeksforGeeks.
- Microsoft Azure Data Fundamentals Certified.
- IBM Skills Build - Getting Started with AI Certified.
- Machine Learning with Python certified from Coursera.
- Maintained CGPA of 8.95.

## Leadership & Extracurriculars

- Core team member of the University Dance Club; managed and organized multiple events. [2024-Present]
- Won the Inter-College women's cricket tournament from our university. [2024]